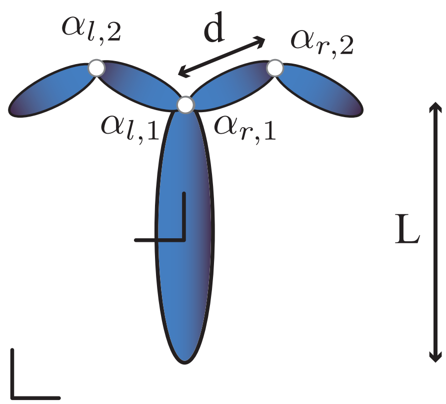
MAE 269: HW 4

Due Friday May 8, 2026 at midnight

1 Gait optimization for a flagellar swimmer



Develop the equations of motion and simulate the locomotion of a flagellar based swimmer as shown above. The center link is L long with a floating base reference frame attached at the center of the link. There are two appendages on the left and right side. Each appendage has two joints and two links that are controllable. Set the joint angles such that when all angles are zero the limbs point straight up along the length of the body. Define the angle directions such that a positive increase in the left joints results in a CCW rotation, and a positive increase in the right joints results in a CW rotation. Thus setting $\alpha_{r,i}(t) = \alpha_{l,i}(t)$ results in symmetric motion of the limbs across the left and right side of the body. If we restrict ourselves to symmetric gaits then the shape space is two dimensional $\alpha_1(t) = \alpha_{r,1}(t) = \alpha_{l,1}(t)$ and $\alpha_2(t) = \alpha_{r,2}(t) = \alpha_{l,2}(t)$.

- Design by hand a gait that will move the robot forward.
- Develop a gait generation function using a truncated Fourier series for α_1, α_2.
- Perform a gait optimization to individually (in different optimization simulations) optimize the quantities:
  1. Maximum forward distance in a cycle
  2. Minimum cost of transport

2 Required and optional paper review

Please provide a short paragraph describing your thoughts on the required reading below. This can be in any form: tell me what you liked, didn't like, understood, didn't understand. There is no wrong answer as long as you read and reflect.

- Diaz, K., Robinson, T. L., Aydin, Y. O., Aydin, E., Goldman, D. I., & Wan, K. Y. (2021). A minimal robophysical model of quadriflagellate self-propulsion. Bioinspiration & Biomimetics, 16(6), 066001.